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Roll No

CE-6004-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Highway Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Assume missing data suitably.

1. a) What are the factors affecting Geometric design.
Explain. 7
b) The speed of overtaking and overtaken vehicles is 80 and 50 kmph respectively. On two way traffic load, the acceleration of overtaking vehicles is 0.99m/sec^2 . Calculate OSD, mention the minimum length of overtaking zone and draw the sketch of the overtaking zone with all details. 7
2. a) Explain any two methods of flexible pavements design. 7
b) Calculate the stresses at interior, edge and corner region of cement concrete pavement using westergaards equation. Use the following data. 7
Wheel load = 5200 kg
pavement thickness = 20 cm
Poissons ratio of concrete = 0.15
subgrade modulus = 6kg/cm^3
Youngs modulus of concrete = $3 \times 10^5 \text{kg/cm}^2$
Radius of contact Area = 15 cm.

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3. a) What is Geotextiles? Describe the functions of Geotextiles in road construction? 7
b) Discuss the requirements of good highway drainage system. 7
4. Explain about:
a) Channelised and unchannelised intersections. 7
b) At grade and grade separated intersections. 7
5. a) Explain the different runway geometrics as recommended by ICAO. 7
b) The length of runway under standard conditions is 600 m. The airport site has an elevation of 100 meters reference temperature is 28°C. If the runway is to be constructed with an effective gradient of 0.5%, determine the corrected runway length. 7
6. a) Explain the elements of airport lighting with neat sketches. 7
b) Explain in detail about airport zoning? 7
7. a) Describe the classification of Highways based on location and function? 7
b) Write short notes on: 7
i) Indian Road Congress (IRC)
ii) Central Road Research Institute (CRRRI)
8. a) Describe the objectives of the engineering surveys and explain the engineering surveys conducted for highway alignment. 7
b) Explain in detail about the crushing test, abrasion test, impact test and soundness test on the aggregates used for highway and road construction. 7

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